Chapter 2 – Polynomial Equations and Inequalities

The Factor Theorem

Factor Theorem

$x - a$ **is a factor of** $f(x)$ **if and only if** $f(a) = 0$

The Remainder Theorem

Remainder Theorem

When $f(x)$ **is divided by** $x - a$ **the remainder is** $f(a)$

factor / solve a higher degree polynomial (n = 3 *or* 4) expression or equation

The division statement is..	2. Divisor – what you are dividing by
Dividend = divisor x quotient + remainder	3. Quotient – what you conclude/solve
1. Dividend – what you start with	4. Remainder – what is remaining

Express equation in factored form, must use factor theorm & synthtic ÷

$f(x) = 9x^4 + 6x^3 - 35x^2 - 24x - 4$

① Find Factors of f(x)

$\frac{b}{a}$ = factors a = first number, b = last number { You must find all factors of (a) and (b), then divide them to find factors of f(x).

(a = 9) – ±1, 3, 9
(b = 4) – ±1, 2, 4

$x = \pm 1, \frac{1}{3}, \frac{1}{9}, 2, \frac{2}{3}, \frac{2}{9}, 4, \frac{4}{3}, \frac{4}{9}$

These are all possible x-values

② Use factor theorm to determine if they are x-values of f(x)

f(1) = −48 ∴ Not factor
f(−1) = −12 ∴ Not factor
f(2) = 0 ∴ Yes a factor!
This makes the equation = 0, it is an x-value

③ Use synthetic division to find the rest of the x-values

divisor

2	9	6	−35	−24	−4
	↓	18	48	26	4 +
x	9	24	13	2	0
	x^3	x^2	x^1	x^0	R

- Put the x-value on the left
- Make sure #s are in decending order according to degree. If there is a missing degree, add a zero in place of it.
- Bring down the first term, and multiply it by the divisor
- add the number by the next term, repeat

Write the new equation

$(9x^3 + 24x^2 + 13x + 2)$

Synthetic division reminder: if divison is a value greater than 1 (Ex. (2x−3) then factor out the two and divide by it after synthetic division has been done. DON'T divide the remainder though.

④ Find the rest of the x-values

Since you are unable to factor by grouping, you must repeat process 2–3 again

$f(-\frac{1}{3})$ = is a factor!

$-\frac{1}{3}$	9	24	13	2
	↓	−3	−7	−2 +
x	9	21	6	0
	x^2	x^1	x^0	R

⑤ Use factoring or quadratic formula

$(9x^2 + 21x + 6)$

$\frac{-21 \pm \sqrt{225}}{18} = x = -1/3,\ x = -2$

⑥ Write Conclusion statement

* remember to include x-values found using factor theorm

$x = \{-\frac{1}{3}, -2, 2\}$ $(9x^4 + 6x^3 - 35x^2 - 24x - 4) = (x-2)(x+2)(3x+1)^2$

lowest to highest write original equation first, then write factors. two x-values so, add squared

Chapter 4 – Rates of Change

ARC / EIRC , h = 0.01 AND word problems

ARC: $\dfrac{f(x_2) - f(x_1)}{x_2 - x_1}$ **EIRC:** $\dfrac{f(a+h) - f(a)}{h}$

A ball is thrown and it's height H(t) is above level ground at time (t) units is modelled by $H(t) = 15t - 4t^2 + 18$. $H(t) = -4t^2 + 15t + 18$ rearranged

From what height is the ball thrown?

Make your $x = 0$

$H(0) = 18m$ ∴ Ball is thrown at 18metres.

When does the ball strike the ground?

set $y = 0$, use quadratic formula ($x \geq 0$)

$0 = -4t^2 + 15t + 18$

$\dfrac{-15 \pm \sqrt{513}}{-8} = \begin{cases} x = 4.71 \text{ sec} \\ x = -0.95 \text{ sec} \end{cases}$

∴ Ball strikes ground at approx 4.71 seconds.

What is the ARC of the ball from 0.72 sec (x_1) → 2.24 sec. (x_2)

Formula

$\dfrac{f(x_2) - f(x_1)}{x_2 - x_1}$

① Plug in x-values in equation

$f(0.72) = 26.7264$

$f(2.24) = 31.5296$

② Plug into equation

$= \dfrac{31.5296 - 26.7264}{2.24 - 0.72}$

$= 3.16$

③ Therefore statement

∴ The ARC from 0.72 sec to 2.24 sec is $3.16 \dfrac{\text{metres}}{\text{sec}}$ (3.16 m/s)

When is the instentanious ROC at 3.47 seconds ($x = a$)

Formula

$\dfrac{f(a+h) - f(a)}{h}$

$h = 0.01$ always

① Plug in values for (a+h) & (a)

$f(a) = 21.8864$

$f(a+h) = 21.7584$

② Plug into equation

$\dfrac{21.7584 - 21.8864}{0.01}$

$= -12.8$

③ Therefore

∴ The EIRC at the height of ball at 3.47s is $-12.8 \dfrac{\text{metres}}{\text{sec}}$.

When does the ball reach a height of 15.56

$15.56 = -4t^2 + 15t + 18$

set equation to $y = 0$ and solve

$0 = -4t^2 + 15t + 2.44$

use quad formula

$\dfrac{-15 \pm \sqrt{264.04}}{-8} = \begin{cases} x = -0.156 \text{ – no negative time} \\ x = 3.906 \end{cases}$

∴ The ball reaches this height at ≐ 3.91 sec

How to find max height

$\max = \dfrac{-b}{2a}$ (sec)

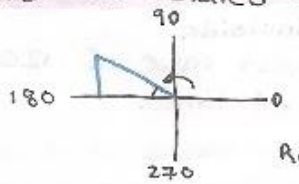
equivalent expressions using the related angle

Related and Co-related angles

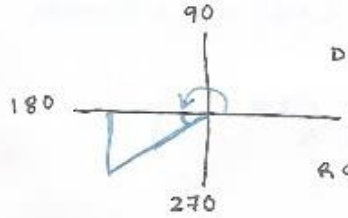
θ - principal angle

R - related angle

C - co-related angle



Degrees	$R = 180 - \theta$ $C = 90 - R$
Radians	$R = \pi - \theta$ $C = \pi/2 - R$



Degree	$R = \theta - 180$ $C = 90 - R$
Radians	$R = \theta - \pi$ $C = \pi/2 - R$

$\theta = 180 + R$

$\theta =$	$R = 360 - \theta$ $C = 90 - R$
$R =$	$R = 2\pi - \theta$ $C = \pi/2 - R$

Equivalent related and co-related

$\sin R = \cos C$
$\sin R = \cos(90 - R)$

$\sec R = \csc C$
$\sec R = \csc(90 - R)$

$\tan R = \cot C$
$\tan R = \cot(90 - R)$

Ex. $\sin \frac{\pi}{8}$ as an equivalent cos expression

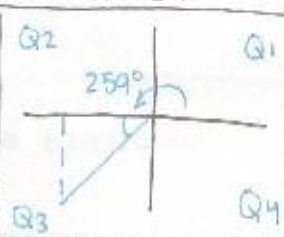
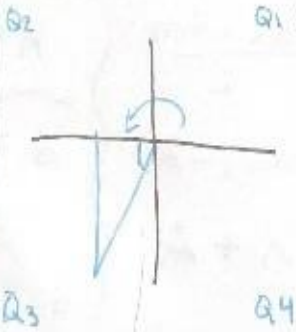
$$\sin \frac{\pi}{8} = \cos\left(\frac{\pi}{2} - \frac{\pi}{8}\right) = \cos \frac{3\pi}{8}$$

① Find principal angle = Related angle

② If principal angle is larger than 180, do $(180 - \theta)$

③ Find principal angle = co-related angle

④ Check answers w/ calculator

Ex.	Sketch	Related ∠	Co-Related ∠
$\tan 259°$ (+) in Q3		$\tan\theta = \tan R$ $\tan 259° = \tan 79°$ $(259 - 180)$	$\tan\theta = \cot C$ $\tan 259 = \cot(90 - 79)$ $\tan 259° = \cot 11°$
$\sec \frac{15\pi}{11}$ (245.45) In Q3 (−sec)		$\sec\theta = -\sec R$ $\sec \frac{15\pi}{11} = -\sec \frac{4\pi}{11}$ $\left(\frac{15\pi}{11} - \pi\right)$	$\sec\theta = -\csc C$ $\sec \frac{15\pi}{11} = -\csc \frac{3\pi}{22}$ $\left(\frac{\pi}{2} - \frac{4\pi}{11}\right)$